Kirti Mahapatra

+91-97761-24300 | kirtimahapatra07@gmail.com | linkedin/kirti-mahapatra | github/kirti-007 | LC: 323 · 190E/124M/9H

Education

Indian Institute of Technology (IIT) Bhilai — June 2025 – Present
M.Tech, Computer Science and Engineering — *Raipur, Chhattisgarh*

Veer Surendra Sai University of Technology (VSSUT), Burla — Nov 2020 – Jun 2024
B.Tech, Computer Science and Engineering | CGPA: 8.88/10 — *Odisha*

Skills

Languages: C/C++, Python, JavaScript, TypeScript, Java, SQL
Frontend: React.js, Next.js, Redux Toolkit, React Query, Tailwind CSS, Framer Motion
Backend & DB: Node.js, Express.js, REST APIs, WebSockets (Socket.io), MongoDB, MySQL, JWT
Data & Tools: NetworkX, Pandas, BeautifulSoup, PyVis, scikit-learn, Git, Vercel, Postman, Linux
CS Fundamentals: DSA | OS | DBMS | Computer Networks | OOP | Graph Theory | **GATE CSE 2025**: 645/1000

Experience

Frontend Developer Intern — Oct 2023 – Jun 2024
Bottom Street — *Remote*

- Rebuilt 12 marketing and product pages from scratch in Next.js 14 with TypeScript, improving Lighthouse Performance score from 54 to 91 and reducing Largest Contentful Paint (LCP) by 38%.
- Migrated all client-side `useEffect` data fetching to Next.js Server Components and SSG, cutting Time-to-First-Byte from 920ms to 310ms and eliminating 18 redundant client-side re-renders on high-traffic landing pages.
- Implemented JSON-LD schema markup and semantic HTML across all 12 pages, contributing to a 30% improvement in organic search ranking for target keywords as measured in Google Search Console over 90 days.
- Built a reusable TypeScript component library (buttons, modals, form inputs with Zod validation) consumed across the entire site, cutting per-feature UI development time by an estimated 4 hours per sprint.

Projects

Global Alloy Ecosystem — Network Science on Materials Data | *Python, NetworkX, Pandas, PyVis* — Jan 2025 | GitHub

- Scraped and structured 500+ alloys and their elemental compositions from public materials databases; modeled the global alloy ecosystem as a weighted graph with 200+ nodes and applied PageRank, betweenness centrality, and Louvain community detection to identify the 10 most supply-chain-critical metals.
- Detected 6 material communities via Louvain modularity algorithm, correlating cluster structure with real-world industrial sectors (aerospace, electronics, construction); built an interactive PyVis force-directed graph enabling drill-down on any element node to expose co-occurrences and criticality score.

CollabNote — Real-Time Collaborative Markdown Editor | *React, TypeScript, Node.js, Socket.io, MongoDB* — GitHub

- Designed and built multi-user real-time document editing with WebSocket (Socket.io) sync and operational-transformation conflict resolution; architected the React/TypeScript frontend using custom hooks (`useEditor`, `useAutoSave`), reducing component re-renders by 60% via `React.memo` and `useMemo`.
- Implemented JWT + refresh-token auth with HTTP-only cookie storage (XSS-safe) and per-document role-based access control (owner / editor / viewer); added compound indexes on `(userId, lastModified)`, reducing document-list fetch latency from ∼400ms to ∼60ms at 1,000 document scale.

AlgoViz — Algorithm Visualization Engine | *React, TypeScript, Tailwind CSS* — GitHub | Live

- Built an interactive browser-based visualizer for 15+ graph and sorting algorithms (Dijkstra, BFS/DFS, A*, Merge Sort, Quick Sort) rendered on an SVG canvas; implemented a custom scheduler using `requestAnimationFrame` and a generator-based execution model to allow pause, rewind, and step-through at 60fps without blocking the UI thread.
- Designed a drag-and-drop graph editor with directed/undirected toggle and edge weight assignment; serialized complete graph state to URL query params, enabling shareable single-link algorithm demos with zero backend.

SnapLink — URL Shortener with Click Analytics | *Node.js, TypeScript, Redis, MongoDB, React* — GitHub | Live

- Engineered Base62/SHA-256 short-code generation sustaining 100 concurrent redirects with p99 latency under 20ms via Redis caching; implemented Redis sliding-window rate limiter (100 req / 15 min per IP) blocking abuse with zero impact on legitimate traffic.
- Built a click analytics pipeline capturing UTM parameters, user-agent, and geolocation per click; aggregated time-series data using MongoDB `$group` and `$bucket` aggregation pipeline, visualized as a live Recharts dashboard.